Invoice Amount Prediction

1.Data Collection:

The dataset contains retail invoice transactions including product details, customer information, and billing values. The target variable is **total invoice amount,** which is calculated using unit price, quantity, and tax. The dataset is useful for sales analysis and prediction models.

2. Data Preprocessing:

The dataset contains:

Total Rows:1000

Total Columns:20

Therfore there is no null values and duplicate values.

3. Univariate & Bivariate analysis:

Exploratory Data Analysis (EDA) is an important step in data analysis where we explore and **visualize** the data to understand its main **features, find patterns** and see how different variables are related.

1. Univariate Analysis

Univariate analysis studies **one variable** at a time to understand its **characteristics** and **distribution.**

- **Histograms:** Show how data values are distributed.
- **Box plots:** Help detect outliers and show data spread.
- **Bar charts:** Used for categorical variables.

2. Bivariate Analysis

Bivariate analysis examines the **relationship** b**etween two variables** to understand how they interact or influence each other.

- **Scatter plots:** Show the relationship between two numerical variables.
- **Correlation coefficient:** Measures the strength of the relationship between variables.
- **Cross-tabulation:** Displays the relationship between two categorical variables.
- **Line graphs:** Compare two variables over time to identify trends.
- **Covariance:** Shows how two variables change together.

4. Feature Selection & model creation:

[13]:

	Linear	SVMl	SVMnl	Decision	Random
f_regression	1.0	1.0	0.663345	0.99998	0.999977

[16]:
```
result
#6
```

[16]:

	Linear	SVMl	SVMnl	Decision	Random
f_regression	1.0	1.0	0.559033	0.999947	0.999974

[19]:
```
result
#5
```

[19]:

	Linear	SVMl	SVMnl	Decision	Random
f_regression	1.0	1.0	0.632066	0.999973	0.999973

Best 5 features

Index(['unit_price', 'quantity', 'vat', 'cogs', 'gross_income'], dtype='object')

Here **SelectKBest** is used to determine the best features by **k values 4, 5, 6**

And therefore **5 features** are selected for model creation as **unit price, quantity, vat, cogs, gross income** and thus Random forest algorithm gives more **accuracy 0.99**

5. Final model:

Here **RandomForestRegressor** is used to create model and using pickle model is created and dump and input is given by user to **predict the invoice amount prediction**

Here the output, invoice amount is predicted based on the **walmart dataset,** therefore follows:

```
Accuracy: 0.9999785554099285
```

[14]:
```
# save the model
filename='Random_model.sav'
pickle.dump(regressor,open(filename,'wb'))
```

[15]:
```
# Load the model
loaded_model=pickle.load(open("Random_model.sav",'rb'))
```

[16]:
```
unitprice_input=float(input("Unit Price:"))
quantity_input=float(input("Quantity:"))
vat_input=float(input("Vat:"))
cogs_input=float(input("Cost of goods sold:"))
gross_input=float(input("Gross Income:"))
```

```
Unit Price: 50
Quantity: 15
Vat: 120
Cost of goods sold: 45
Gross Income: 500
```

[18]:
```
# Result
Amt_Prediction=loaded_model.predict([[unitprice_input,quantity_input,vat_input,cogs_input,gross_input,]])
```

[19]:
```
print("Invoice Amount Prediction:",Amt_Prediction)
```